MIL-S-52098B
2 August 1971
SUPERSEDING
MIL-S-52098A
11 October 1963

MILITARY SPECIFICATION

STIFFENER ASSEMBLY, BALK CONNECTING

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the stiffener assembly and its components for support of the deck balk superstructure.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Box, Fiberboard.
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple-Wall.

Military

MIL-T-704	- Treatment and Painting of Materiel.

STANDARDS

Federal

FED. STD. No. 595	- Colors.

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

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DRAWINGS

ME

DL13208E9871 - Stiffener Assembly, Balk Connecting.

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN WELDING SOCIETY

A5.10 - Aluminum and Aluminum-Alloy Welding Rods and Bare Electrodes.
B3.0 - Standard Qualification Procedure.

(Application for copies should be addressed to the American Welding Society, 345 East 47th Street, New York, N. Y. 10017.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, N. Y. 10017.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification.

(Application for copies should be addressed to the Uniform Classification Committee, Tariff Publishing Officer, Room 202, Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification.

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street NW, Washington, D. C. 20036.)

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3. REQUIREMENTS

3.1 Description. The stiffener assembly and its components shall be as shown on Data List DL13208E9871 and as specified herein.

3.1.1 Drawings. The drawings forming a part of this specification are engineering design drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the supplier or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.2 First article (preproduction model). The supplier shall furnish one or more stiffener assemblies or components, as applicable, for examination and testing within the time frame specified to prove prior to starting production that his production methods will produce stiffener assemblies or components that comply with the requirements of this specification. Examination and tests shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.3).

3.3 Material. Material shall be as specified herein and as shown on the applicable drawings. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification.

3.4 Welding electrodes. Welding electrodes shall conform to AWS A5.10, classification ER4043, size as required.

3.5 Identification marking. The stiffener assembly or components, as applicable, shall be metal-die stamped as shown on the applicable drawing. Die stamping shall be legible after painting.

3.6 Treatment and painting. The stiffener assembly or components, as applicable, except pins and screw threads shall be cleaned, treated, and painted in accordance with MIL-T-704, type B, except that the color number designation shall be 34087 in accordance with FED. STD. No. 595. Contact surfaces, regardless of whether such contact is between similar or dissimilar metals, shall be cleaned, treated, and painted prior to assembly.

3.7 Stenciling. The nomenclature, Federal stock number, weight, and cube shall be stenciled on each stiffener assembly. The letters "NET WT" shall precede the numerals in the weight and the letters "CU" shall precede

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the cubic displacement. Markings shall be Gothic-type capitals and Arabic numerals 3/4 inch high. Stenciling shall conform to MIL-T-704.

3.8 Workmanship.

3.8.1 Aluminum fabrication. Before any work or layout is done, aluminum shall be straightened by methods that will not result in injury to the metal. Care shall be used in handling aluminum during fabrication to avoid excessive scratching or damage to edges. Finished members shall be free from laminations, kinks, bends, scratches, or winds. Corners shall be square and true. Aluminum 3/8-inch thick and over shall be cut by routing or sawing. Aluminum under 3/8-inch thick may be cut by shearing. Aluminum shall not be cut by methods using heat.

3.8.2 Riveting. Components shall be bolted in position before riveting. Jigs or frames shall be used to maintain alinement and tolerance. Holes for countersunk heads shall be of such depth and angle that the head shall fill the hole and be flush with the surrounding surface. Heads of driven rivets shall be uniform in shape and concentric with the rivet shank axis, and shall have full bearing on the surfaces of the joined members. Rivets shall fill the holes in the joined members. Caulking or redriving of rivet heads shall not be permitted. Rivet tension shall not be created by driving in incorrectly fitted members. A rivet with a head deficient in size, not uniform in shape, offcenter, loose, or misshapen in any way shall be considered defective. Defective rivets shall be removed without injury to the joined metal. Rivets shall be driven cold.

3.8.3 Welders and welding.

3.8.3.1 Welders. Before assigning any welder to manual welding work covered by this specification, the supplier shall provide the contracting officer with certification that the welder has passed qualification tests for any of the aluminum alloys as prescribed by either of the following listed codes for the material joined and the type of welding operation to be performed and that such qualification is effective as defined by the particular code:

Standard Qualification Procedures of the American Welding Society (B3.0)
Welding Qualifications of the ASME Boiler and Pressure Vessel Code (Section IX).

3.8.3.2 Welding. The surfaces of parts to be welded shall be free from scale, paint, grease, and other foreign matter. Welding shall be done by the shielded metal-arc, gas metal-arc (MIG), or gas tungsten-arc (TIG)

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process. Aluminum welding process shall conform to the provisions of Part B, Section IX (Welding Qualifications) of the ASME Boiler and Pressure Vessel Code. All welded parts shall be free from cracks and other imperfections that may reduce the effectiveness of the part. Work shall be positioned for flat welding when practicable. Before welding over previously deposited weld metal, all traces of slag shall be removed and the deposit and adjoining base metal shall be wirebrushed clean.

3.8.4 Interchangeability. Stiffener assemblies shall be fabricated to be duplicates within the tolerances specified and shall be interchangeable.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The supplier is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.1.2 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the Supplementary Quality Assurance Provisions (SQAP) shown on the drawings. The drawings specify the characteristics requiring SQAP inspection, the sampling plan, and the basis for acceptance and rejection (see 6.4 and 6.5).

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).
(c) Inspection of preparation for delivery (see 4.6).

4.3 Preproduction inspection.

4.3.1 Examination. The preproduction stiffener assembly or component(s), as applicable, shall be examined as specified in 4.4.2.1 and for the defects specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

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4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.4.2 Examination.

4.4.2.1 Individual. Each stiffener assembly shall be inspected for conformance to the dimensions covered by the supplementary quality assurance provisions (SQAP) shown on the drawings under "Major - 100 percent inspection". Nonconformance to any of these dimensions shall constitute failure of this examination and shall be cause for rejection.

4.4.2.2 Samples. Samples selected in accordance with 4.4.1 shall be examined for the major and minor defects specified in 4.5.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.5 Inspection procedure.

4.5.1 Examination. The stiffener assembly or component, as applicable, shall be examined for the following defects:

Major

101. All dimensions designated major on the SQAP other than those under 100 percent inspection not as specified.
102. Material not as specified.
103. Components missing or not as specified.
104. Workmanship not as specified.
105. Weld defects such as flux inclusions, spatter, undercutting, porosity, incorrect size, or lack of fusion with parent metal.
106. Riveting not as specified.

Minor

201. All dimensions not designated major on the SQAP not as specified.
202. Identification marking incorrect, missing or illegible.
203. Treatment and painting not as specified.
204. Stenciling not as specified.
205. Cadmium plating not as specified.

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4.6 Inspection of preparation for delivery.

4.6.1 Quality conformance inspection of pack.

4.6.1.1 Unit of product. For the purpose of inspection a completed pack prepared for shipment shall be considered a unit of product.

4.6.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

107.	Materials and containers not as specified for level A or B. Each incorrect material or container shall constitute one defect.
108.	Gross weight of wood boxes exceeds 200 pounds for level A or B.
109.	Strapping not zinc coated for level A.
110.	Gross weight or size of fiberboard boxes exceeds the limitations of the applicable box specification for level B.
111.	Marking missing, illegible, incorrect, or incomplete.

5. PREPARATION FOR DELIVERY.

5.1 Packing. Packing shall be level A, B, or C as specified.

5.1.1 Level A. Each completely assembled (with pins in place) stiffener assembly shall be prepared for shipment loose.

5.1.1.1 Components and parts. When the pin, spring, and spring holder assembly are procured separately from the stiffener assembly, the items shall be packed in close-fitting wood boxes conforming to PPP-B-601, overseas type, style optional. The gross weight of each box shall not exceed 200 pounds and the boxes shall be closed and strapped in accordance with the appendix to the box specification. Strapping shall be zinc coated.

5.1.2 Level B. Each completely assembled (with pins in place) stiffener assembly shall be prepared for shipment loose.

5.1.2.1 Components and parts. When the pin, spring, and spring holder assembly are procured separately from the stiffener assembly, the items shall be packed in close-fitting wood boxes conforming to PPP-B-601, domestic type, style optional; in close-fitting fiberboard boxes conforming to PPP-B-640, class 2, style optional; or PPP-B-636, V3c, V11c, V13c, or V15c, style optional. The gross weight of each wood box shall not exceed 200 pounds and the boxes shall be closed and strapped in accordance with the

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appendix to the applicable box specification. The gross weight or size of fiberboard boxes shall not exceed the limitation of the applicable box specification. Box closure and strapping shall be in accordance with the appendix to the applicable box specification.

5.1.3 Level C. Each completely assembled (with pins in place) stiffener assembly shall be prepared for shipment loose. Components and parts, when procured separately from the stiffener assembly, shall be packed to assure carrier acceptance and safe delivery to destination at lowest ratings in fiberboard or wooden containers complying with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.2 Marking. Shipping containers and loose assemblies shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The stiffener assembly is intended for use as a beam for support of the balk superstructure and as a means of connecting the balk superstructure to the supporting floats.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Time frame required for submission of the preproduction model and number of assemblies or components, as applicable, required (see 3.2).
(c) Level of packing required (see 5.1).

6.3 Preproduction model. Any changes or deviations of production stiffener assemblies or components from the approved preproduction model during production will be subject to the approval of the contracting officer. Approval of the preproduction model will not relieve the supplier of his obligation to furnish stiffener assemblies or components conforming to this specification.

6.4 Supplementary quality assurance provisions (SQAP). The contracting officer should require the supplier to maintain records of all SQAP inspections. A suggested paragraph is as follows:

"The supplier shall maintain complete records of all examinations and tests performed to verify the requirements of classified SQAP characteristics. The records shall include as a minimum, lot size,

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sample size, drawing requirement, actual measurement, number and type of deficiencies found, quantity approved, quantity rejected, corrected action taken when applicable."

6.5 Definitions.

6.5.1 Supplementary quality assurance provisions (SQAP). A SQAP is a contractual requirement that supplements section 4 of the specification. SQAP indicates the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

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Preparing activity:

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User activity:

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-S-52098B - Stiffner Assembly, Walk Connecting

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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